

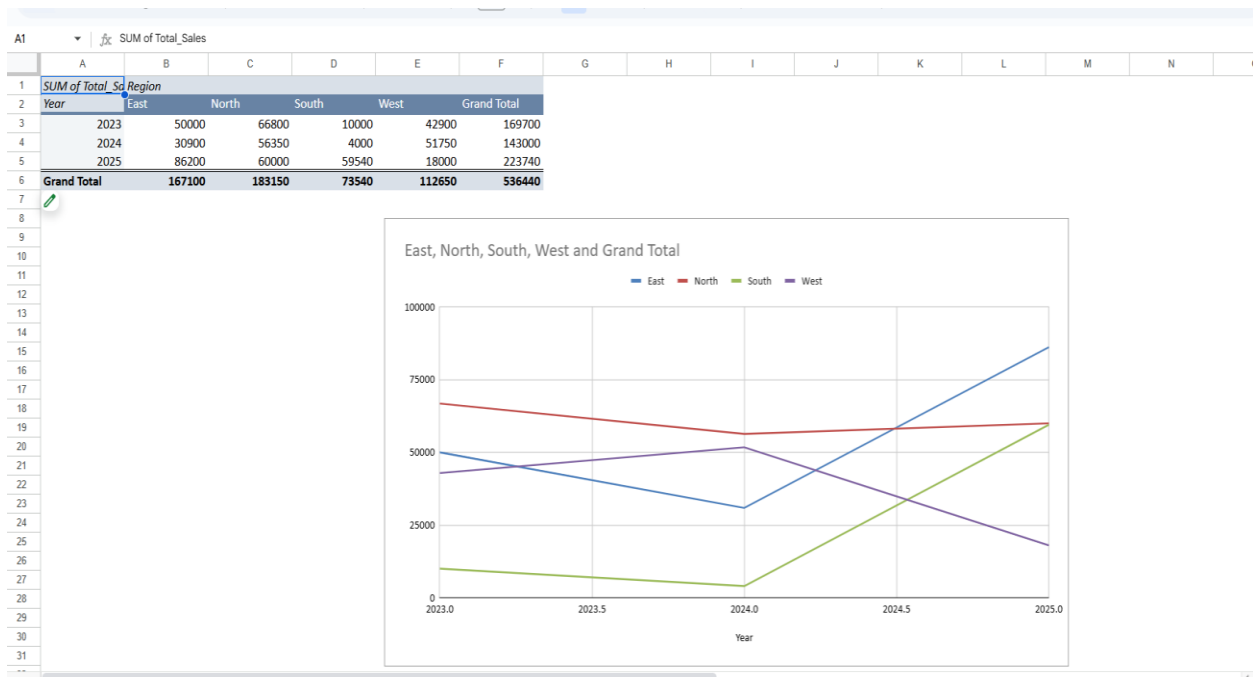
GOOGLE SHEET LINK:

https://docs.google.com/spreadsheets/d/1nMUX7RSPjedmUunKVfeoA2drIQO1yFrZLE_tJvYTMuQ/edit?usp=sharing

ANS TO THE QUES OF PART-A

TASK 1: REGIONAL SALES SUMMARY

1. Overview: The dataset was analyzed to determine total sales performance across different regions (East, North, South, West) from 2023 to 2025.
2. Regional Performance Based on the Grand Total sales volume:
 - Top-Performing Region: North (Total Sales: 183,150)
 - Lowest-Performing Region: South (Total Sales: 73,540)
3. Trend Analysis:
 - Overall Trend: Sales dipped in 2024 (143,000) but recovered significantly in 2025 (223,740).
 - Regional Volatility: The South region showed a massive growth spike, jumping from negligible sales in 2023/2024 to nearly 60,000 in 2025. Conversely, the West region saw a sharp decline in 2025.
 - Stability: The North region remained the most consistent performer across all years.
4. Visualization:

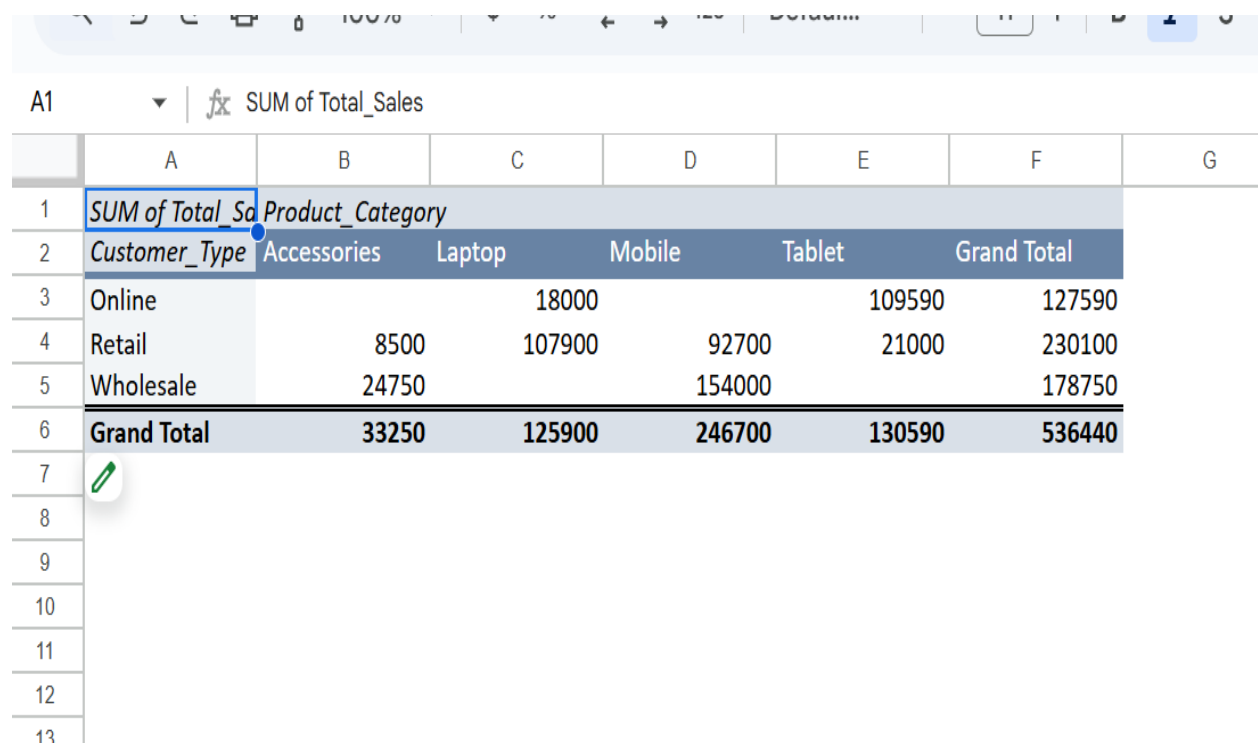


Task 2: Customer and Category Insights

1. Highest Contributing Customer Type : Retail customers contributed the most to total revenue, generating 230,100 in sales. This outperforms Wholesale (178,750) and Online (127,590) channels.

2. Popular Categories by Segment

- Retail: The dominant category was Laptops.
- Wholesale: Sales were driven primarily by Mobiles.
- Online: The most popular category was Tablets.



The screenshot shows an Excel spreadsheet with a pivot table. The formula bar at the top displays 'SUM of Total_Sales'. The pivot table is structured with 'Customer_Type' as the row labels and 'Product_Category' as the column labels. The data is summarized in the following table:

Customer_Type	Accessories	Laptop	Mobile	Tablet	Grand Total
Online		18000		109590	127590
Retail	8500	107900	92700	21000	230100
Wholesale	24750		154000		178750
Grand Total	33250	125900	246700	130590	536440

Task 3: Formula Application

1. SUMIFS() – Total sales for Laptops sold in 2024

- Formula: =SUMIFS(I:I, E:E, "Laptop", B:B, 2024)
- Result: 22,500
- Explanation: Calculates the sum of Total_Sales only where the Product Category is "Laptop" AND the Year is "2024".

2. **AVERAGEIFS()** – Average sales for Wholesale customers

- Formula: =AVERAGEIFS(I:I, D:D, "Wholesale")
- Result: 29791.67
- Explanation: Calculates the average of Total_Sales strictly for transactions where Customer_Type is "Wholesale".

3. **COUNTIFS()** – Number of transactions in the East region

- Formula: =COUNTIFS(C:C, "East")
- Result: 6
- Explanation: Counts the number of rows where the text "East" appears in the Region column.

4. **IF()** – Label sales as 'High' or 'Low'

- Formula: =IF(I2 > 30000, "High", "Low")
- Result: (Applied to column)
- Explanation: Checks each transaction row. If Total_Sales is greater than 30,000, it returns "High"; otherwise, it returns "Low".

5. **VLOOKUP()** – Display Product Category given a Name

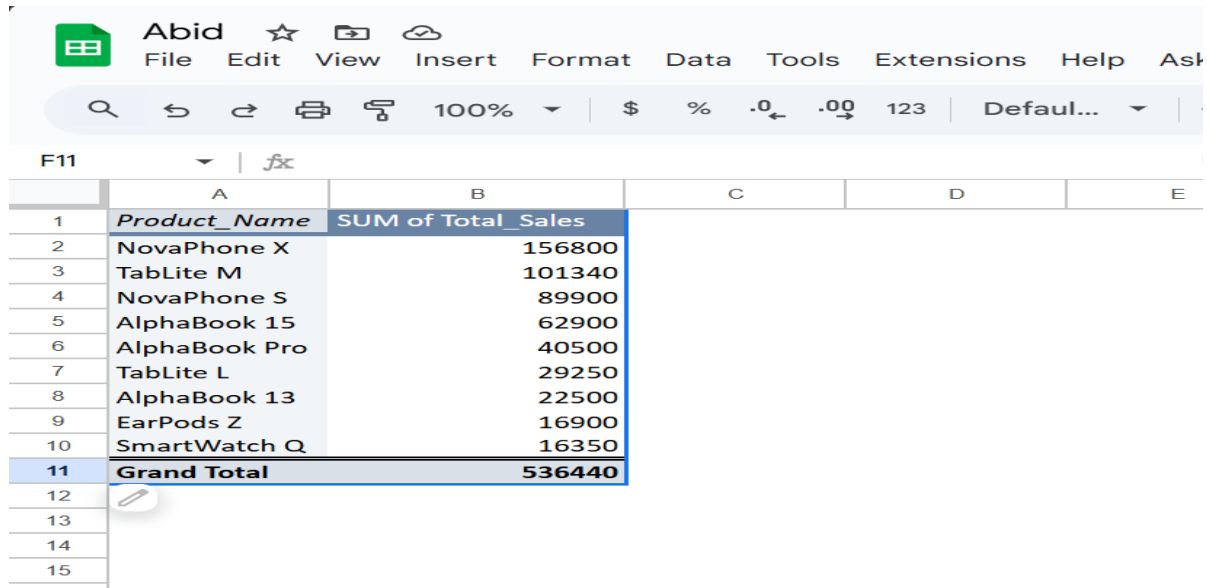
- Formula: =VLOOKUP("NovaPhone S", P:Q, 2, FALSE)
- Result: Mobile
- Explanation: Searches for ("NovaPhone S" in a helper table and returns the Category from the second column.

6. **XLOOKUP()** – Find the Total Sales of Laptops

- Formula: =XLOOKUP("Laptop", E:E, I:I)
- Result: 20,000
- Explanation: Searches the Product_Category column for "Laptop" and returns the Total_Sales from the same row. Note: It returns the value of the first match found (Transaction T001).

Task 4: Product Ranking

1. Summary Table:



	A	B	C	D	E
1	Product_Name	SUM of Total_Sales			
2	NovaPhone X	156800			
3	TabLite M	101340			
4	NovaPhone S	89900			
5	AlphaBook 15	62900			
6	AlphaBook Pro	40500			
7	TabLite L	29250			
8	AlphaBook 13	22500			
9	EarPods Z	16900			
10	SmartWatch Q	16350			
11	Grand Total	536440			
12					
13					
14					
15					

2. Observation : The NovaPhone X is the clear market leader with 156,800 in sales. The data indicates a strong customer preference for high-value hardware (Mobiles and Tablets). Low-cost accessories like EarPods Z and SmartWatch Q are the lowest performers, suggesting they are not significant revenue drivers for the company.

ANS TO THE QUES OF PART B: REGIONAL STRATEGY

1. Identification: The region with the lowest sales is the South, with a total of 73,540.

2. Strategic Recommendations:

- Strategy 1: Shift Product Focus to High-Value Hardware. Data analysis reveals that the South struggled in 2023/2024 when sales were driven by low-cost Accessories. The 2025 spike was due entirely to Mobile and Tablet sales. We should discontinue marketing accessories in the South and allocate the entire budget to stocking NovaPhones and TabLites to maintain the 2025 momentum.
- Strategy 2: Expand Retail Partnerships. Our customer segmentation analysis (Part A, Task 2) proves that Retail is the most profitable channel. Since the South's recent success was driven by retail transactions, the sales team should prioritize acquiring new Retail partners in the South region rather than relying on Online sales, which have historically been weaker there.

ANS TO THE QUES OF PART C: THEORY & CONCEPTS

1. Data Summarization: Data summarization is the process of condensing large volumes of raw, granular data into meaningful insights using statistical measures like counts, totals, and averages. It effectively turns "noise" into "signal." In business analytics, raw data often consists of thousands of transaction rows which are impossible to interpret individually. Summarization is crucial because it allows stakeholders to instantly spot patterns, outliers, and performance trends (e.g., identifying the North as the top region). Without summarization, data remains unstructured and unusable for strategic planning.

2. Two advantages and one limitation of Excel summarization tools

Advantages:

1. **Accessibility:** Excel tools like Pivot Tables are user-friendly and do not require coding knowledge (unlike Python or SQL), allowing employees at all levels to derive insights.
2. **Dynamic Interactivity:** Tools like slicers and charts are linked dynamically to the data source. If a sales figure is updated, the entire summary report updates instantly, allowing for real-time analysis.

Limitation: Scalability: Excel struggles with "Big Data." As datasets grow to hundreds of thousands of rows, summarization tools can become slow or unstable compared to dedicated database management systems.

3. Data summarization helps managers make data-driven decisions: Data summarization replaces intuition with evidence. Instead of guessing which products to stock, a manager can view a summarized ranking (like Task 4) and see empirically that Mobiles generate 3x the revenue of Laptops. This leads to efficient resource allocation; managers can cut funding for underperforming regions (like the West in 2025) and reinvest in growing markets. Furthermore, historical summaries allow for accurate forecasting, helping managers optimize inventory levels to prevent stockouts and maximize profit margins based on proven trends rather than gut feelings.